

Problem

Using only two op amps, design a circuit to solve

$$-v_{\text{out}} = \frac{v_3 - v_1}{5} + \frac{v_1 - v_2}{2}$$

Step-by-step solution

Step 1 of 3

The output voltage required is ,

$$V_{\text{out}} = \frac{V_1 - V_3}{5} + \frac{V_2 - V_1}{2}$$

Using only two op-amps we can obtain this output by taking inverting configuration of two op-amps. To get $\frac{V_2 - V_1}{2}$ this voltage as output of second op-amp chooses resistors and voltage sources in negative manner for 1st op-amp and 2nd op-amp.

Draw the equivalent circuit as shown in Figure 1.

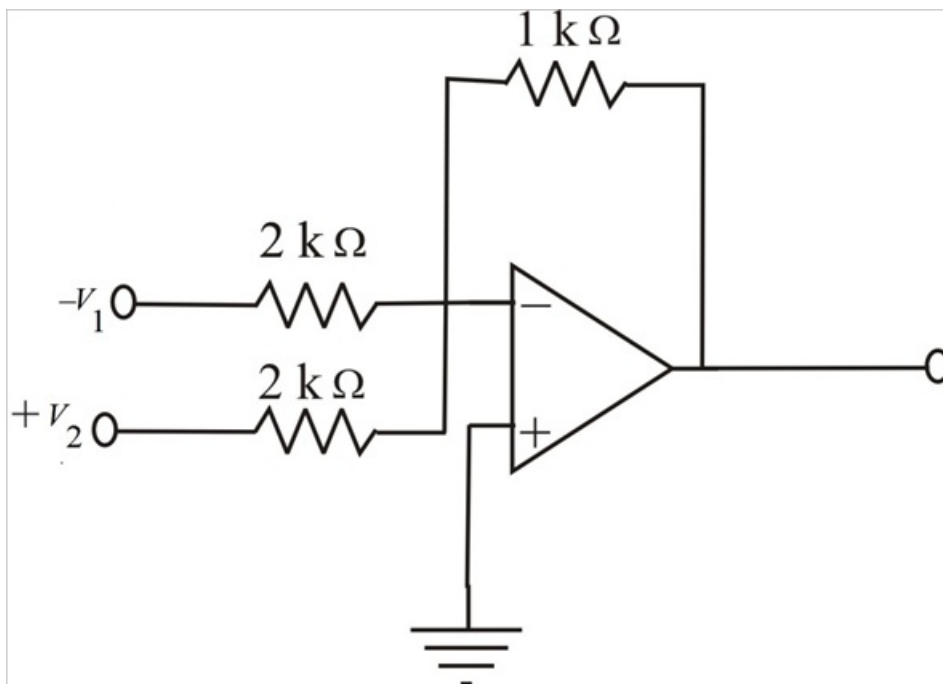


Figure 1

Step 2 of 3

The output voltage of 1st op-amp is,

$$\frac{-1}{2}(-V_1) - \frac{1}{2}(V_2) = \frac{V_1 - V_2}{2}$$

Connect the second op amp as shown in Figure 2.

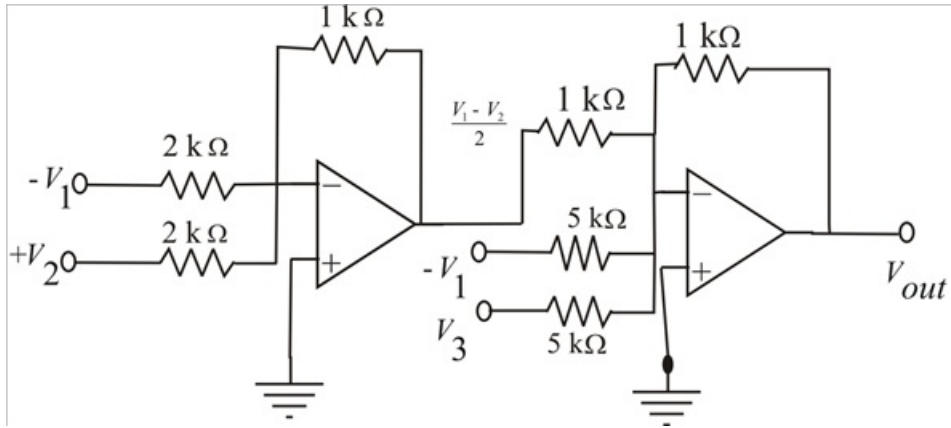


Figure 2

Step 3 of 3

The output voltage is,

$$\begin{aligned} V_{out} &= \frac{-1}{1} \left(\frac{V_1 - V_2}{2} \right) - \frac{-1}{5} (V_3) - \frac{-1}{5} (-V_1) \\ &= \frac{V_2 - V_1}{2} + \frac{1}{5} (V_3) + \frac{1}{5} (V_1) \\ -V_{out} &= - \left(\frac{V_2 - V_1}{2} \right) - \left(\frac{V_1 - V_3}{5} \right) \\ &= \left(\frac{V_1 - V_2}{2} \right) - \left(\frac{V_1 - V_3}{5} \right) \end{aligned}$$

Hence the required out is obtained using the two op-amps.